Limits and continuity
MTH 201 - Business Calculus

Phan Xuân Thành

Faculty of Mathematics and Informatics
Hanoi University of Science and Technology
e-mail: thanh.phanxuan@hust.edu.vn



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Outline

1. Limit of functions
2. Continuity of functions

Limits

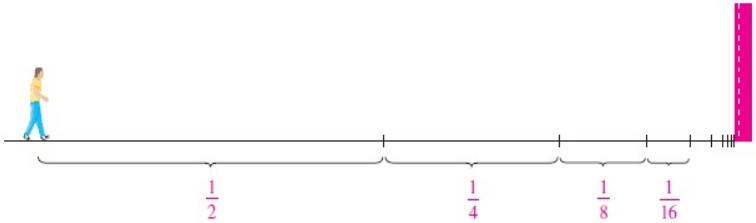
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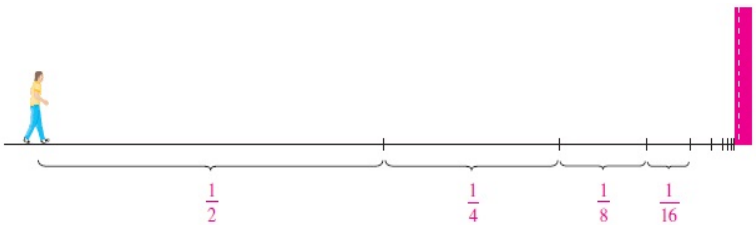
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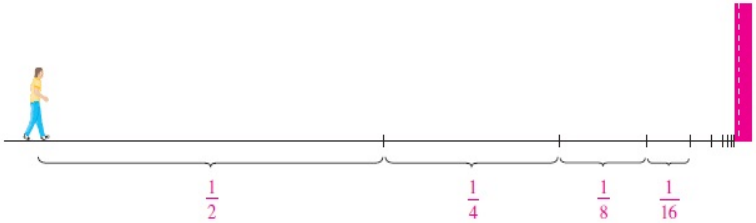
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This process can always be continued and can never be ended.

We would say that the limit of the distance between the man and the wall is zero.

Limit of functions

Definition

As x approaches a, the limit of $f(x)$ is L, written $\lim_{x \to a} f(x) = L$, if all values of $f(x)$ are close to L for all values of x that are sufficiently close, but not equal, to a. The limit L must be a unique real number.

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Example 1. Find $\lim_{x\to 1}(3x + 2)$, $\lim_{x\to 2} \dfrac{x^3 - 8}{x^2 - 4}$, $\lim_{x\to 0} x \sin \dfrac{1}{x}$.

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Theorem

As x approaches a, the limit of $f(x)$ is L if the limit from the left exists and the limit from the right exists and both limits are L. That is,

$$\textit{if } \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L, \textit{ then } \lim_{x \to a} f(x) = L.$$

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Example 2. Given $f(x) = \begin{cases} 2-x, & \text{if } x \geq 1, \\ 2x+1, & \text{if } x < 1. \end{cases}$ Find $\lim_{x\to 3} f(x)$. What is the limit $\lim_{x\to 1} f(x)$?

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Example 3. Given $g(x) = \begin{cases} 1+x, & \text{if } x \leq -1, \\ x^2, & \text{if } -1 < x < 3 \\ 9, & \text{if } x \geq 3. \end{cases}$ Find $\lim_{x\to 3} g(x)$, $\lim_{x\to 2} g(x)$ and $\lim_{x\to -1} g(x)$.

Limits involving infinity

Infinite limits

$$\lim_{x \to a} f(x) = \infty \;\Leftrightarrow\; \forall M > 0, \exists \delta > 0 \text{ such that } f(x) > M \text{ if } 0 < |x - a| < \delta.$$

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Example 4. Let $f(x) = \dfrac{1}{x-3}$. Find

$$\lim_{x\to 2} f(x), \quad \lim_{x\to 3^+} f(x), \quad \lim_{x\to 3^-} f(x), \quad \lim_{x\to 3} f(x)?$$

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Example 5. Find the limits

$$\lim_{x\to 0^+} (x+2)e^{1/x}, \quad \lim_{x\to 0^-} (x+2)e^{1/x}.$$

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Limits at infinity

Sometimes we need to determine limits when the inputs get larger and larger without bound, that is, as the inputs approach infinity:

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Example 7. Find the limits at infinity of the following functions

$$f(x) = \frac{5}{x-1}, \quad g(x) = \frac{2|x|}{x+3}, \quad h(x) = \frac{3x+1}{\sqrt{x^2+4}},$$

$$p(x) = \frac{2^x - 5}{3^x + 4}, \quad q(x) = \sin x.$$

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Algebraic limits

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Note that for any rational function f, with $a \in D_f$, then $\lim\limits_{x\to a} f(x) = f(a)$.

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Example 8. Find the limit $\lim\limits_{x\to 2} \dfrac{\sqrt{x^2+12}-4}{x-2}$, $\lim\limits_{x\to 1} \dfrac{x-1}{x^n-1}$, $\lim\limits_{x\to 0} \dfrac{\sqrt[5]{1+3x}-1}{x}$.

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Example 9. Given $\lim\limits_{x\to 1} \dfrac{f(x)-5}{x-1} = 4$. Find $\lim\limits_{x\to 1} f(x)$.

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Exercise. Evaluate the limits

$$\lim_{x\to 0}\frac{9^x-5^x}{4^x-3^x}, \qquad \lim_{x\to\infty}\left(\frac{x+2}{x+1}\right)^{x+1}, \qquad \lim_{x\to\infty}\left(\frac{x+1}{x+3}\right)^{2x}.$$

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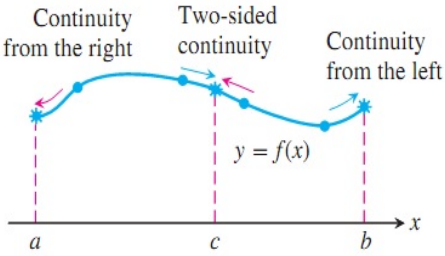
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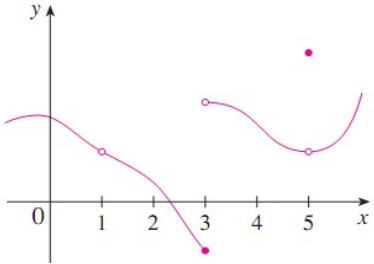
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Exercise 1. The Copy Shoppe charges \$0.08 per copy for quantities up to and including 100 copies. For quantities above 100, the charge is \$0.06 per copy. If x represents the number of copies, the price function is $p(x) = \begin{cases} 0.08x & \text{for } x \leq 100, \\ 0.06x & \text{for } x > 100. \end{cases}$ Is $p(x)$ continuous at $x = 100$?

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Exercise 2. Investigate the continuity of

a) $f(x) = \begin{cases} \dfrac{\sqrt{1+2x}-1}{x} & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$

b) $g(x) = \begin{cases} x^3 - 3x & \text{if } x \geq 1, \\ a - 4x & \text{if } x < 1. \end{cases}$

Average rates of change

If a car travels 110 mi in 2 hr, its *average rate of change (speed)* is 110 mi/2 hr, or 55 mi/hr (55 mph). Suppose you accelerate on a freeway and, glancing at the speedometer, you see that at that instant your *instantaneous rate of change* is 55 mph. These are two quite different concepts. The first you are probably familiar with. The second involves ideas of limits and calculus.

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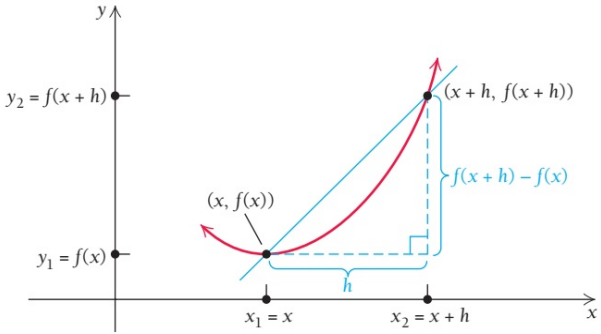
The average rate of change of y with respect to x, as x changes from x_1 to x_2, is the ratio of the change in output to the change in input:

$$\frac{y_2 - y_1}{x_2 - x_1}, \quad \text{where } x_1 \neq x_2.$$

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Difference quotient as average rates of change

Let $y = f(x)$. we write $x_1 = x$ and $x_2 = x + h$



$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x+h) - f(x)}{x + h - x} = \frac{f(x+h) - f(x)}{h}$$ difference quotient.

Example. Find the simplified difference quotient for each function listed.

a) $f(x) = x^2$ b) $f(x) = \dfrac{1}{x^2}$ c) $f(x) = \sqrt{x}$.